

Fixed Geometry, Not Learned Residuals: Exposing Ego-Motion Shortcuts in Sensor-Free Pedestrian Trajectory and Intent Prediction

Submission report · Anonymous · blind review

Generated August 30, 2026 · PIE + JAAD · protocol $\tau=16$, $\eta=45$, TTE 30–60

Abstract. Egocentric pedestrian tracks confound pedestrian motion with camera motion. Zhang et al. decompose the two using ego-speed as input and soft supervision; LIM deletes speed to avoid causal confusion. We ask whether classical offline geometry can recover camera motion without any speed sensor, and whether a learned module can reproduce that geometry from tracks alone. We introduce a standing-pedestrian variance gate and show that three learned LEMC variants (unsupervised, OBD-magnitude, ORB-regression) all fail it, while full-affine ORB compensation evaluated at each agent's image position passes (93.1% PIE, 96.9% JAAD). Fixed ORB is trajectory-neutral on PIE (ADE 60.5 vs 60.8 px) and trajectory-improving on JAAD (62.7 vs 83.1 px, −25%). Intent AUC rises monotonically with ego-motion information available (compensated $0.80 < \text{baseline } 0.89 < \text{GT speed } 0.93$), suggesting a substantial share of reported intent accuracy reflects an ego-motion shortcut.

Three linked findings: (1) Fixed ORB passes the physical variance gate; learned residuals do not. (2) JAAD closes the sensor-free gap (−25% ADE). (3) Intent accuracy tracks available ego-motion signal, not pedestrian-specific reasoning alone.

I. Introduction & Method

A camera on a moving vehicle makes every tracked object's image-plane trajectory a mixture of pedestrian motion and camera motion. Zhang et al. use ego-speed as input and soft supervisor; LIM removes speed to avoid causal confusion. Neither approach runs natively on JAAD, which ships no continuous ego-motion signal.

Fixed ORB-affine (main method). Offline ORB + RANSAC partial affine on consecutive frames with pedestrian boxes masked. Transition target at centre (c_x, c_y) is $M[c_x, c_y, 1]^T - [c_x, c_y]$ — not global translation. Translation-only ORB flattened ~21% of standing+moving windows; per-agent affine ~89% (diagnostic, $n=300$).

Learned LEMC (ablation). Multi-agent displacement statistics $\rightarrow$ GRU $\rightarrow$ cumsum residual, with three supervision regimes: none, OBD magnitude, ORB regression.

Variance gate. For standing pedestrians while the vehicle moves: $\text{Var}(\text{compensated}) < \text{Var}(\text{raw})$ over the observation window. Independent of downstream labels.

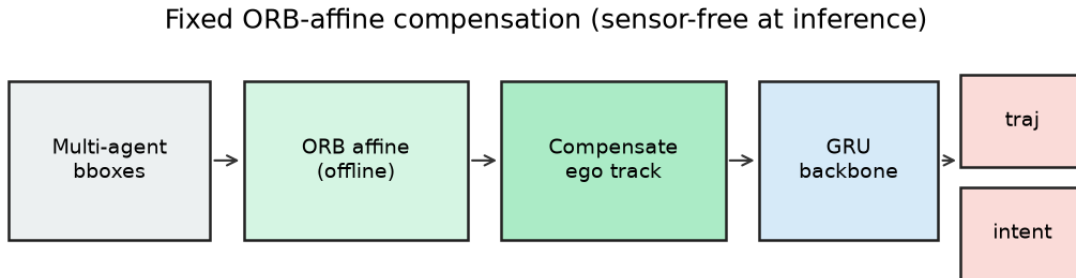


Figure 1. Fixed ORB-affine pipeline: offline geometry compensates ego track before GRU prediction (sensor-free at inference).

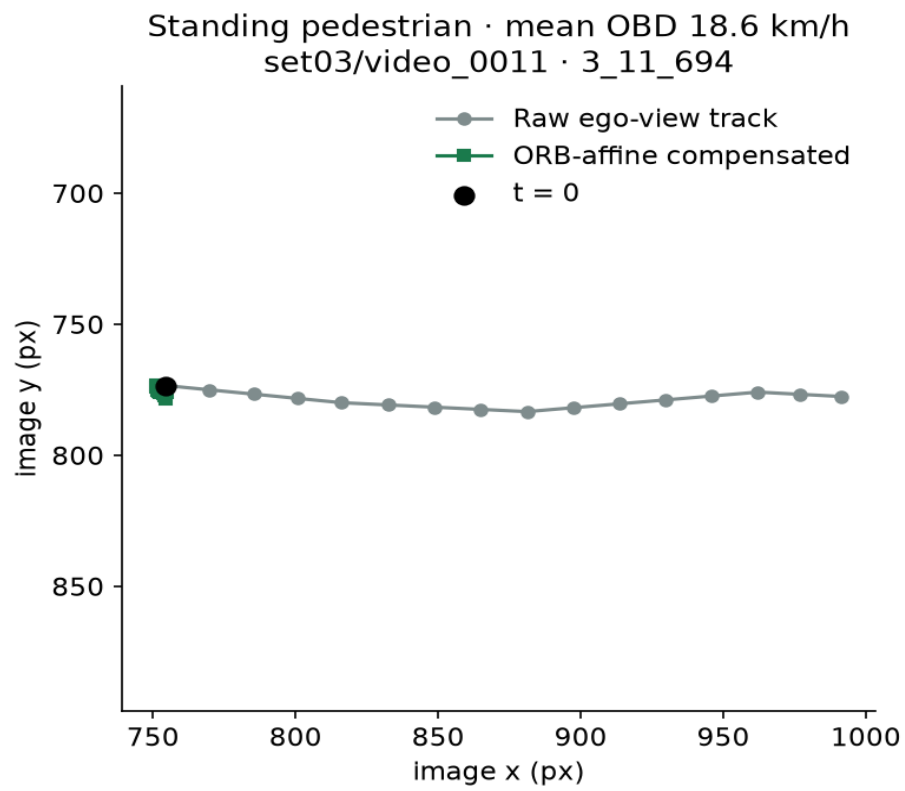


Figure 2. Standing pedestrian (mean OBD 18.6 km/h): raw ego-view track drifts; ORB-affine compensated track stays near-stationary.

II. Experimental Setup

Setting	Value
Observation τ / prediction η	16 / 45 frames
Time-to-event	30–60 frames, 50% overlap
PIE test windows	1,773 (augmented store, median 8 agents/frame)
JAAD test windows	295 (124 scenes with ORB affines)
Backbone	Linear embed $\rightarrow$ 2-layer GRU $\rightarrow$ traj + intent heads
Seeds	0, 1, 2 (mean $\pm$ std)

PIE multi-agent scene · set03/video_0004 · 20 agents



Figure 3. PIE multi-agent scene: annotated bounding boxes used for track-based prediction (ego in orange).

JAAD multi-agent scene · video_0135 · 24 agents

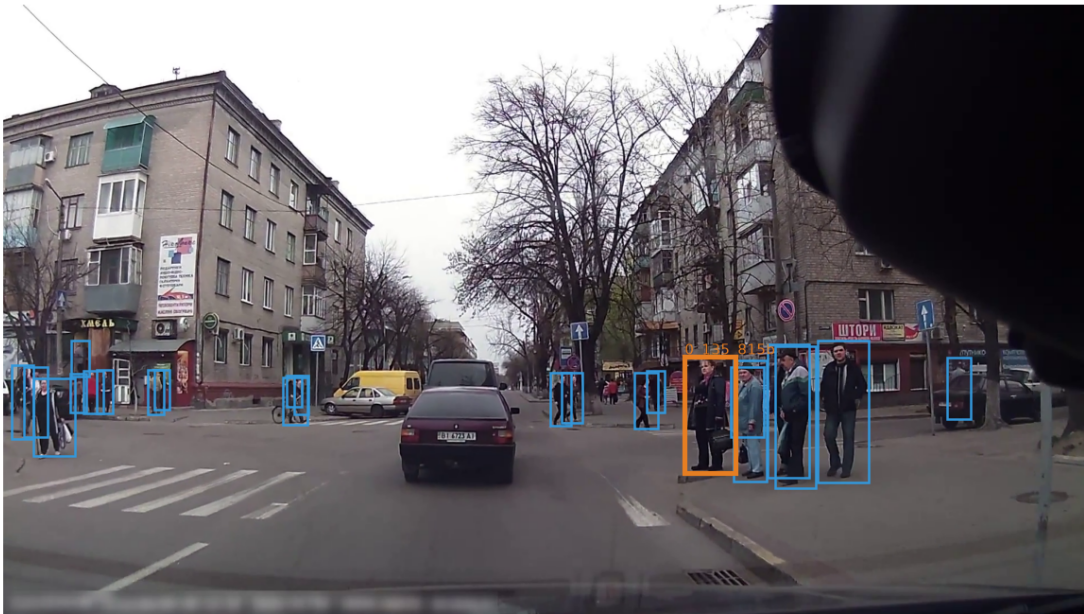


Figure 4. JAAD multi-agent scene: same bbox-only protocol, no OBD/GPS available.

III. Results — PIE

A. Main grid

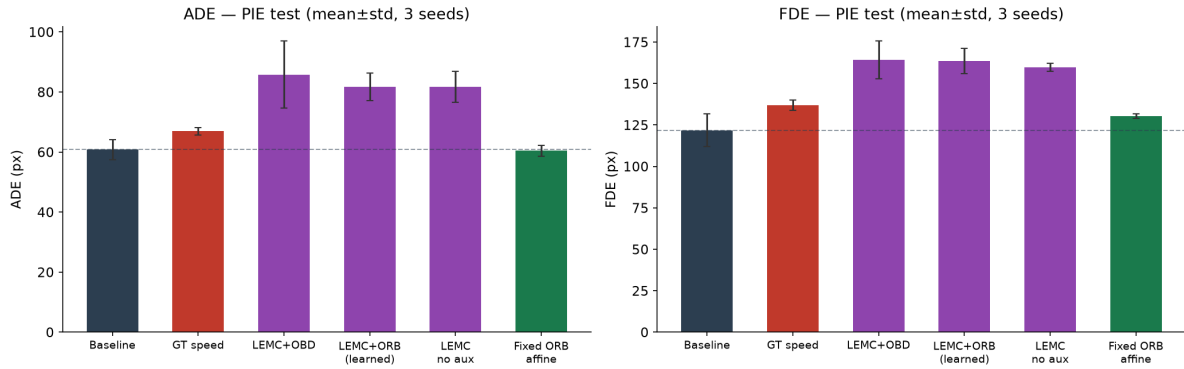


Figure 5. PIE test ADE and FDE (mean $\pm$ std, 3 seeds). Fixed ORB matches baseline; all learned LEMC variants are worse.

Method	ADE	FDE	AUC	F1
Baseline (bbox-only)	60.81±3.31	121.75±9.87	0.89±0.01	0.74±0.03
GT OBD speed	66.96±1.28	136.89±3.02	0.93±0.01	0.85±0.02
LEMC, no aux.	81.68±5.13	159.91±2.39	0.77±0.05	0.64±0.01
LEMC + OBD aux.	85.77±11.19	164.25±11.33	0.80±0.01	0.60±0.03
LEMC + ORB-reg.	81.73±4.61	163.54±7.55	0.81±0.01	0.63±0.00
Fixed ORB affine	60.49±1.83	130.28±1.31	0.80±0.01	0.60±0.01

B. Standing-pedestrian variance gate

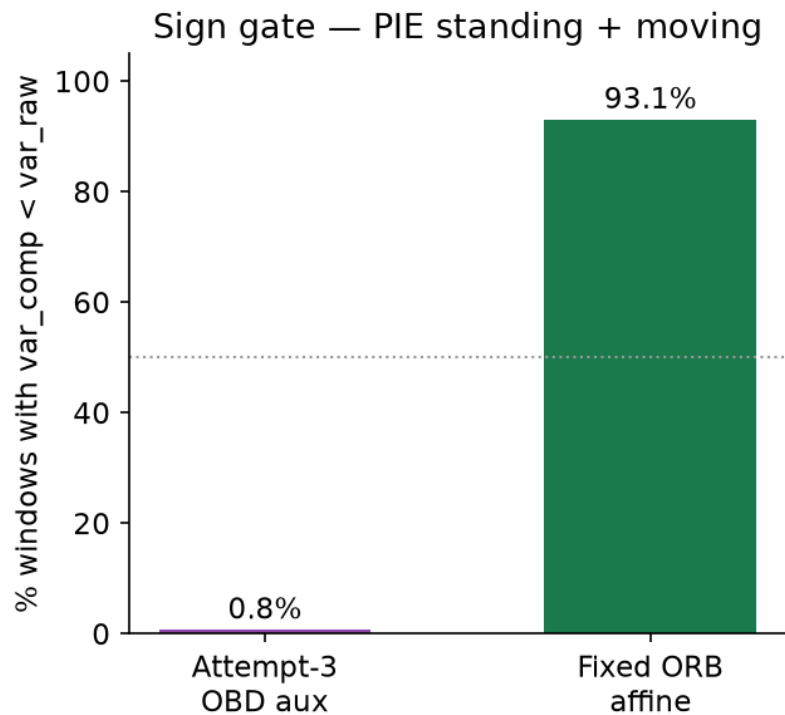


Figure 6. Sign gate on PIE (standing + moving): learned LEMC+OBD 0.8% vs fixed ORB 93.1%.

Method	Dataset	All windows	Standing + moving
LEMC + OBD aux.	PIE	0.7%	0.8% — FAIL
Fixed ORB affine	PIE	76.7%	93.1% — PASS
Fixed ORB affine	JAAD	71.9%	96.9% — PASS

C. ORB geometric validation

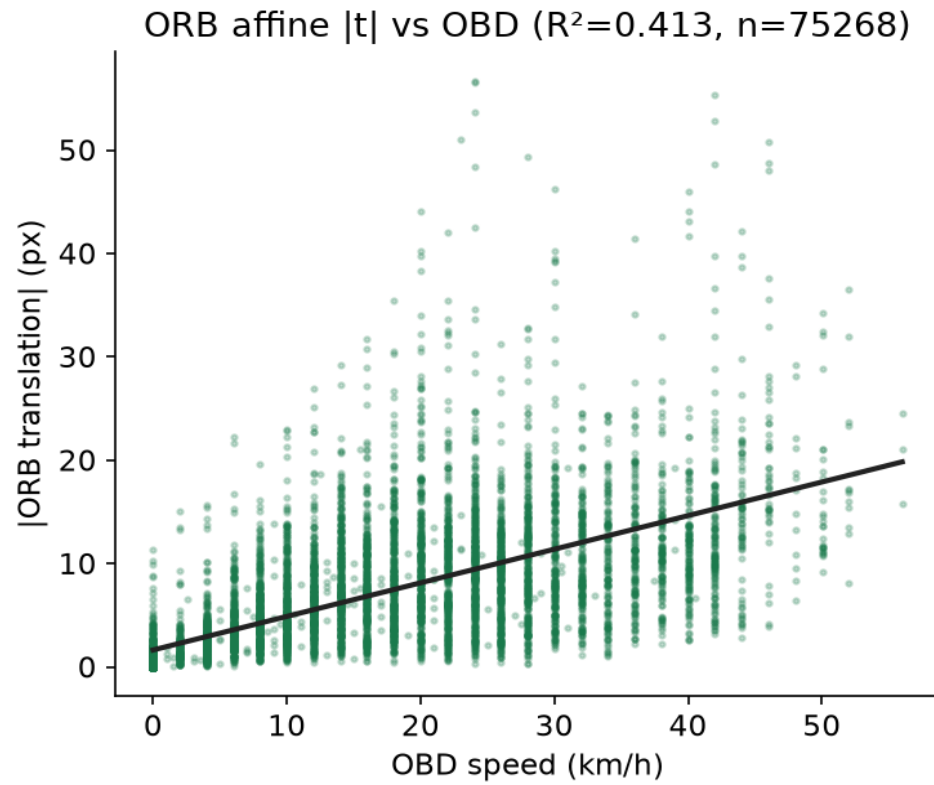


Figure 7. ORB $|t|$ vs OBD speed ($R^2=0.41$, $n=75,268$). Independent sanity check — ORB never consumes speed.

D. Stratified by ego-vehicle state

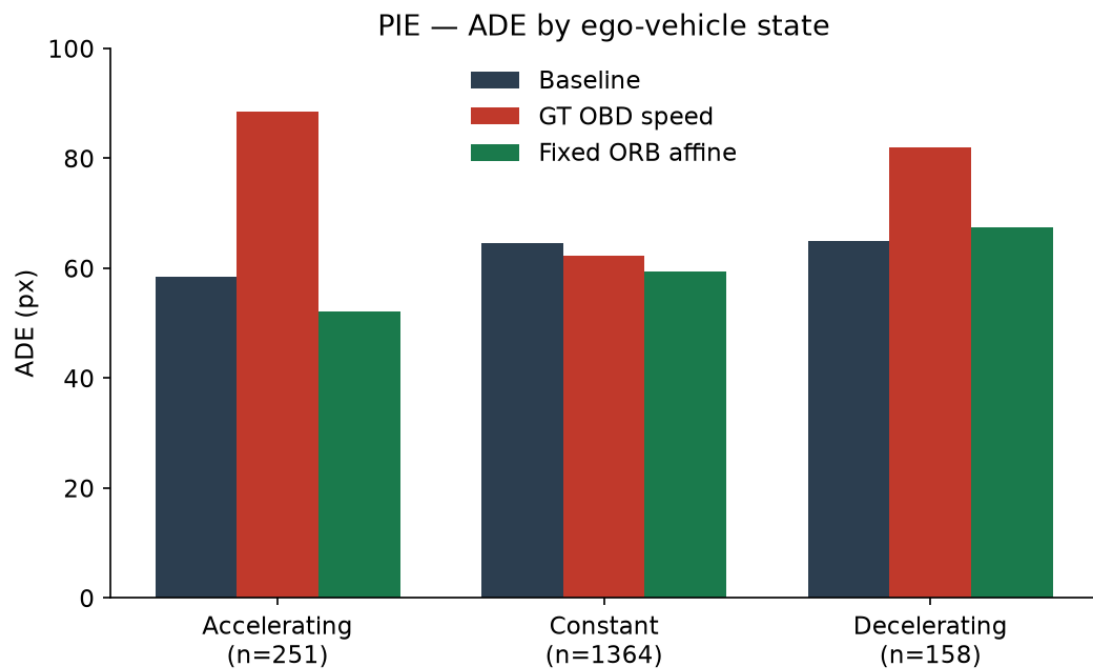


Figure 8. ADE by ego state: fixed ORB best when accelerating; GT speed worst when accelerating/decelerating (LIM-aligned).

Ego state	n	Baseline	GT speed	Fixed ORB
Accelerating	251	58.5 / 127.8	88.6 / 173.2	52.2 / 122.6
Constant	1364	64.5 / 130.6	62.2 / 120.3	59.4 / 130.1
Decelerating	158	65.0 / 139.8	82.0 / 174.7	67.5 / 151.5

IV. Results — JAAD & Intent

A. Cross-dataset trajectory

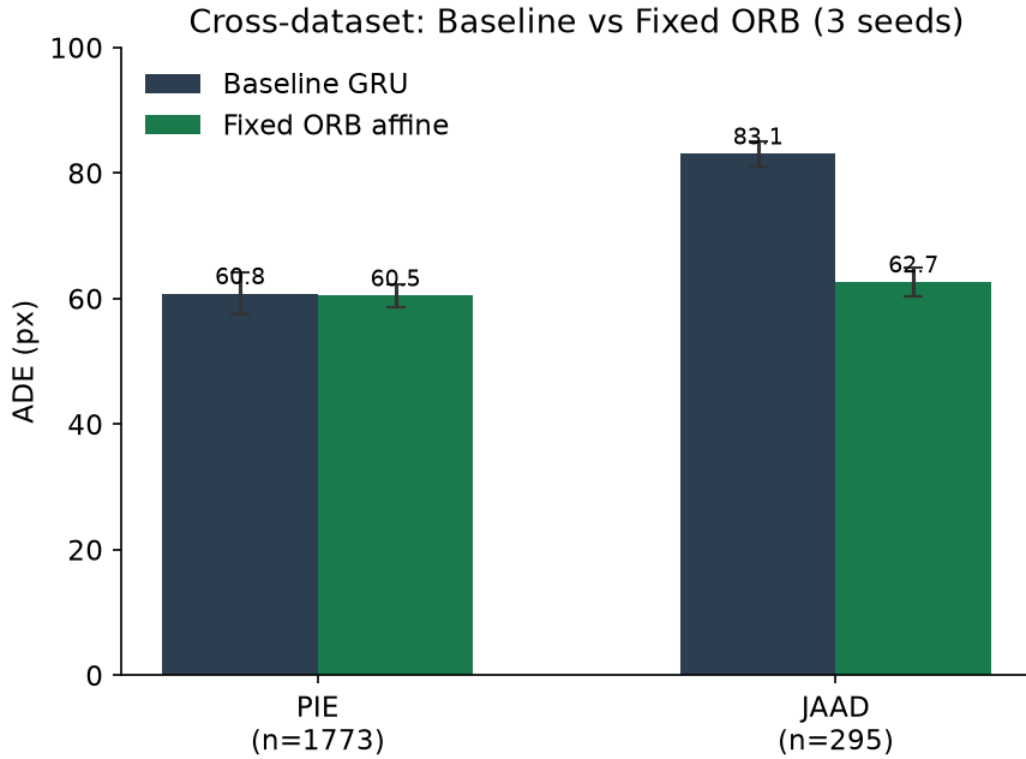


Figure 9. Cross-dataset ADE: PIE neutral (60.8→60.5), JAAD -25% (83.1→62.7).

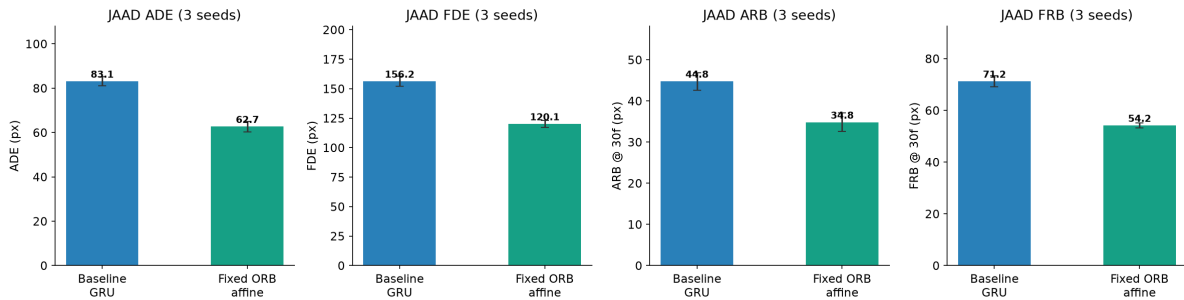


Figure 10. JAAD test metrics (3 seeds): ADE, FDE, ARB, FRB all improve with fixed ORB.

Method	ADE	FDE	ARB	FRB	AUC	F1
Baseline	83.10±2.00	156.25±4.31	44.76±2.15	71.23±2.08	0.47±0.05	0.90±0.00
Fixed ORB	62.68±2.34	120.11±2.82	34.80±2.25	54.17±0.91	0.44±0.00	0.90±0.00

JAAD intent sits at or below chance (AUC 0.47 / 0.44); high F1 (0.90) reflects majority-class imbalance. Contribution on JAAD is trajectory-only.

B. Trajectory overlays on real frames

set03/video_0008 · 3_8_551 · ADE base=323px orb=62px



Figure 11. PIE: baseline vs fixed ORB predictions on a real frame (observed, GT, predicted).

video_0150 · 0_150_970b · ADE base=391px orb=45px



Figure 12. JAAD: same comparison — fixed ORB closer to GT future trajectory.

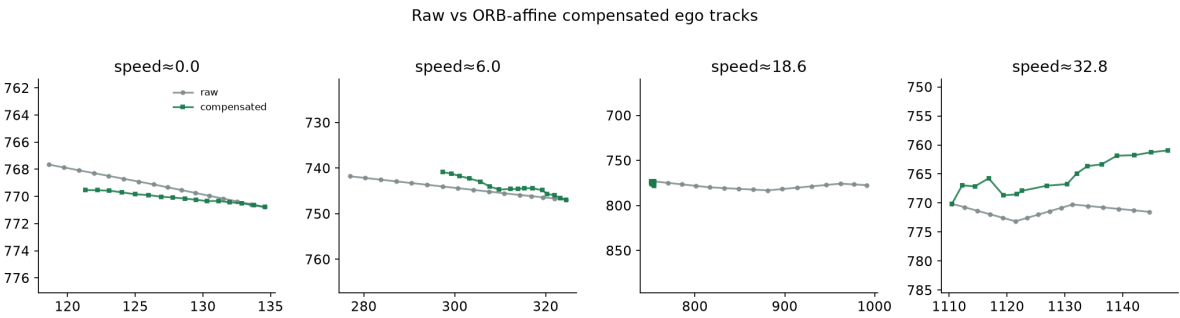


Figure 13. Raw vs ORB-compensated ego tracks across ego-speed percentiles.

C. Intent: ego-motion shortcut

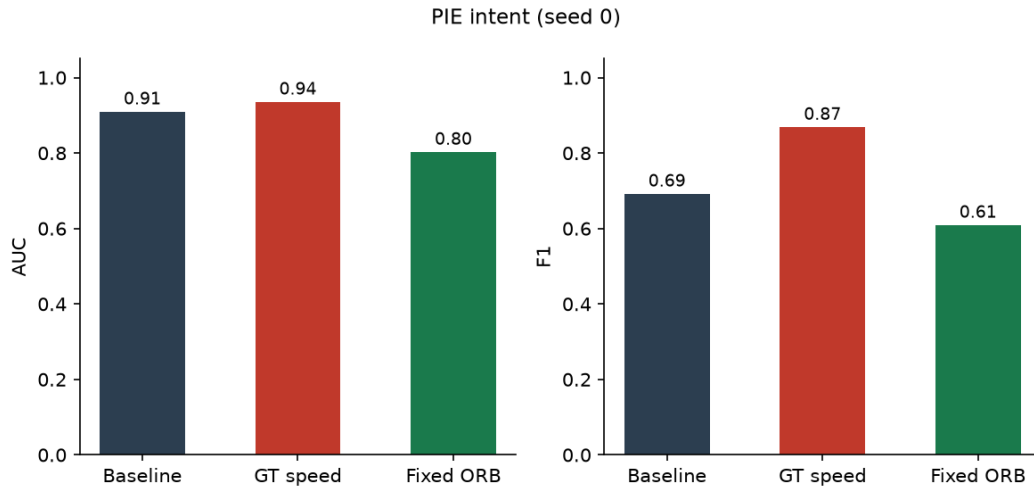


Figure 14. PIE intent AUC and F1 by method: GT speed best, baseline middle, compensated lowest.

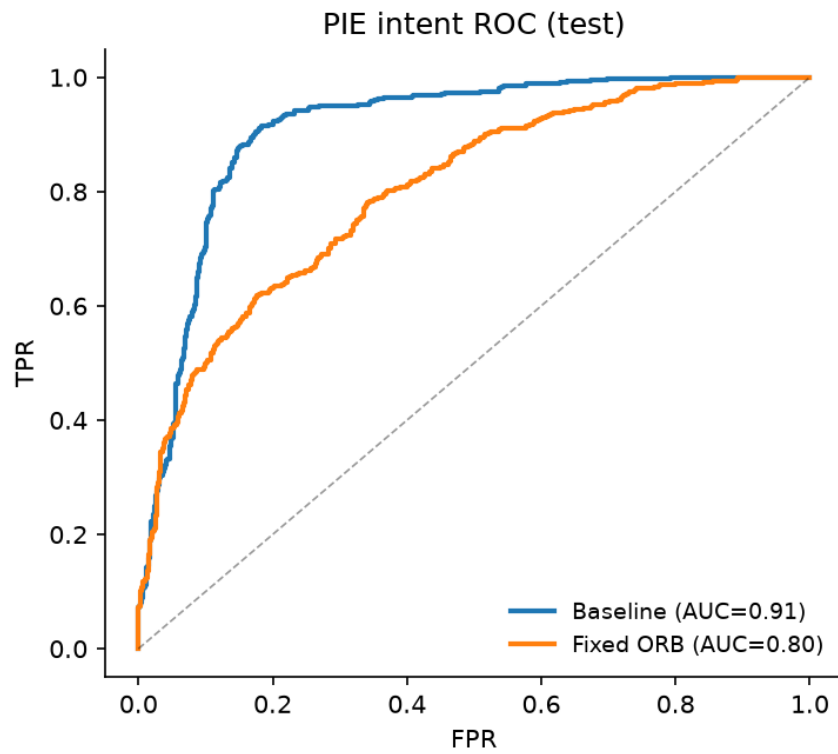


Figure 15. PIE intent ROC: baseline vs fixed ORB (test set).

Ego-motion shortcut pattern: AUC rises monotonically with ego-motion information: fixed ORB (0.80) < baseline (0.89) < GT speed (0.93). F1 follows the same order (0.60 < 0.74 < 0.85). Consistent with LIM's causal-confusion account.

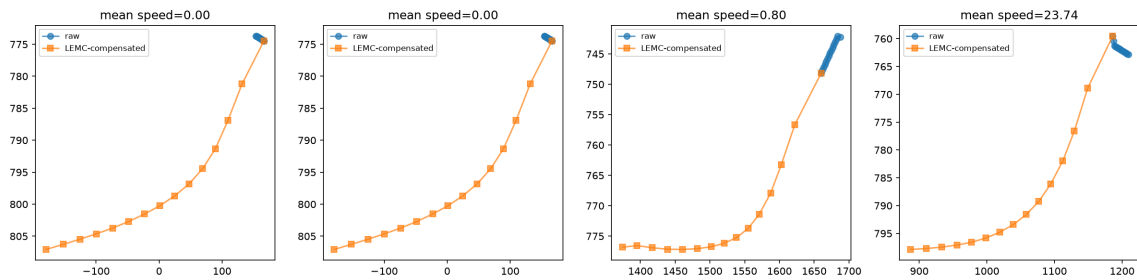


Figure 16. Learned LEMC+OBD: compensated track shows non-physical loops (failed gate).

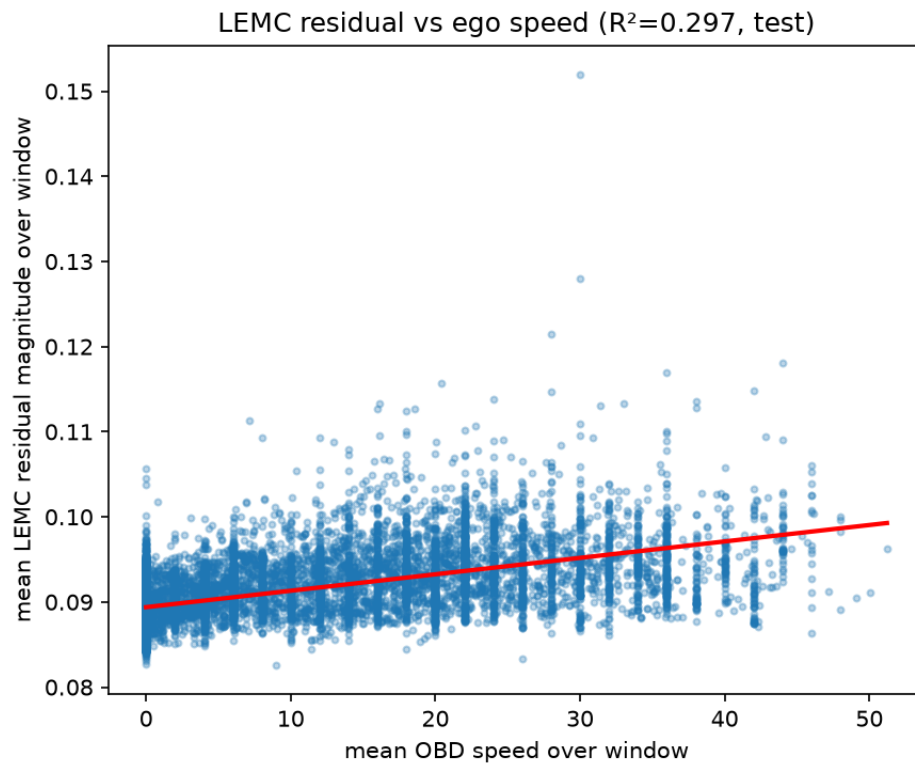


Figure 17. Learned LEMC residual vs OBD: superficial $R^2=0.30$ correlation does not imply valid compensation.

V. Discussion & Conclusion

Why learned residuals fail. A single shared 2D shift cannot represent rotation/scale components of camera motion whose apparent effect depends on image position. Multi-agent statistics average over a target that is not actually shared — structural mismatch, not insufficient data. ORB-regression supervision does not close the gap (81.7 vs 60.5 px), confirming architectural limitation.

Limitations. Bbox-only GRU backbone (not comparable to Zhang I3D ADE 17.41). ORB depth confound. Hold-last for missing future affines. JAAD intent at chance. Ego-motion-shortcut finding from grid observation, not dedicated ablation.

Conclusion. Position-dependent ORB affine passes a physical variance test that three learned alternatives fail. Trajectory-neutral on PIE, −25% ADE on JAAD, zero sensor dependence. Intent AUC tracks available ego-motion signal — sensor-free geometric compensation is viable for trajectory; intent remains open.

Reproducibility

```
python scripts/09b_extract_orb_affine.py → python scripts/12_fixed_orb_affine.py --seed 0 → bash  
scripts/10_run_grid.sh → python scripts/19_make_paper_figures.py → python scripts/generate_submission_report.py
```

Figures: results/paper_figures/ · Checkpoints: checkpoints/ · Configs: configs/

References

- [1] Rasouli et al., PIE, ICCV 2019.
- [2] Kotseruba et al., JAAD 2.0.
- [3] Zhang et al., Decouple ego-view motions, IEEE TIP 2024.
- [4] LIM, Causal confusion in crossing intention, IEEE TITS 2025.
- [5] Rublee et al., ORB, ICCV 2011.
- [6] Fischler & Bolles, RANSAC, CACM 1981.